Thirty-four generations of divergent selection for 8-week body weight in chickens

Tina Flisar,[1] Špela Malovrh, Dušan Terčič, Antonija Holcman, and Milena Kovač

Department of Animal Science, Biotechnical Faculty, University of Ljubljana, Groblje 3, SI-1230 Domžale, Slovenia

ABSTRACT Chickens of the Slovenian commercial Prelux-bro line were divergently selected over 34 generations for high and low BW at 8 wk of age. The aim of the study was to estimate responses to selection with a nonlinear model. Estimates of BW for each generation were provided by the mixed model. For fitting generation means against generation or cumulative selection differential, an exponential model was used. Estimates of realized heritability over generations were derived from regression of the response on cumulative selection differential. After 34 generations, the lines differed by approximately 2,220 g for males and 1,860 g for females. Estimates for a selection limit in the high line were 2,598.4 and 2,144.1 g, for males and females, respectively. A selection limit was not reached in the low line. Half of the selection response was obtained after approximately 6 to 8 generations in the high line and 20 to 28 generations in the low line. Estimated realized heritability decreased over generations. Heritability was larger for females than males and reduction of heritability was more rapid in the high line than in the low line. Genetic SD decreased over generations. Phenotypic SD increased over generations in the high line, but was constant in the low line in the initial 22 generations and decreased thereafter. According to the good fit of the nonlinear model and informative parameter estimates, the results confirmed the usefulness of the nonlinear model for analyzing responses to long-term selection.

Key words: divergent selection, body weight, direct response, nonlinear regression, chicken

2014 Poultry Science 93:16–23
http://dx.doi.org/10.3382/ps.2013-03464

INTRODUCTION

Rapid growth has been the primary emphasis of the majority of breeding programs, particularly in poultry. For studying the genetic and physiological basis and consequences of selection pressure on metric traits, divergently selected lines are important tools. The most important results of long-term selection are estimates of genetic parameters and causes of selection limits. They are also a valuable resource for studies in other disciplines of biology.

Although the effect of selection is reflected in changes of generation means, change of the genetic architecture occurs in the background genome. More than 30 generations of selection are needed to answer most of the questions about genetics of the trait and problems of long-term selection pressure (Eisen, 1980). Only a few long-term selection experiments for BW have been conducted in chickens, mostly at 8 wk of age. The most extended experiments are the Virginia selection experiment (Siegel, 1962; Dunnington and Siegel, 1985; Liu et al., 1994; Dunnington and Siegel, 1996; Márquez et al., 2010, Dunnington et al., 2013) and the Penn State experiment (Barbato, 1992). In Slovenia, a selection experiment in Slovenian commercial Prelux-bro line chickens began at the test station of the Department of Animal Science (Biotechnical Faculty, University of Ljubljana) in 1979 (Terčič and Holcman, 2008; Terčič et al., 2009; Dahmane Gošnak et al., 2010).

Response to selection in short-term selection experiments often shows a linear increase and is therefore estimated from the slope of a regression line fitted to generation mean (James, 1965; Falconer and Mackay, 1996). Response to long-term selection does not continue indefinitely, but slowly diminishes as a result of reducing genetic variance and reaches a "plateau" or selection limit. Polynomial regressions have been applied by Dempster et al. (1952) and James (1965). Dunnington et al. (2013) recently reported a quadratic response over 54 generations of selection for low BW in the Virginia selection experiment, indicating a plateau in the latest generations. The selection limit occurred due to physiological problems. However, the increase of response in the high BW line was linear due to continued genetic variance and beneficial mutations.

©2014 Poultry Science Association Inc.
Received July 4, 2013.
Accepted September 22, 2013.
[1]Corresponding author: tina.flisar@bf.uni-lj.si

Downloaded from http://ps.oxfordjournals.org/ at University of Bern on September 15, 2014

Response to long-term selection can also be described by an exponential regression curve (James, 1965; Richardson et al., 1968; Harris, 1982; Bünger and Herrendörfer, 1994). The advantages of using nonlinear models are 3 easily interpretable parameters, namely selection limits, initial values, and half-life selection. Exponential models also allow estimation of realized heritabilities at a given generation from the slope of the response curve (Frahm and Kojima, 1966). Despite these advantages, only a few experiments were analyzed with exponential models (Bünger and Herrendörfer, 1994; Bünger et al., 1998; Bünger and Hill, 1999), which have not yet been used with poultry data.

The aim of the current study was to estimate the response to selection for BW over 34 generations and heritability for BW by the nonlinear model. The response to selection was studied through genetic parameters. The half-life of the selection process was also obtained.

MATERIALS AND METHODS

Genetic Stocks and Husbandry

The foundation stock for this experiment was established in 1979 from a commercial heavy sire line of Slovenian provenance called Prelux-bro. Two lines were founded from the base population by selecting 10 males and 50 females that were the heaviest at 8 wk of age and those that were the lightest. Within each line, the high-weight males and females were mated at random to establish a high-weight line, and the low-weight males and females were used similarly to establish a low-weight line. After formation in the first generation of selection, the lines were closed, with parents for subsequent generations chosen as the extreme weight males and females from each of the lines. The experiment was conducted for 34 generations. In each line, approximately 50% of the females and 10% of the males were intended to be selected as parents of the next generation. Only healthy and appropriately developed animals, which showed normal behavior, were selected. Therefore, pure truncation selection was not used, especially in the latest generations. To keep the cumulative inbreeding in the 33 generations below 0.3, the minimal number of females (150) and males (50) should be provided according to Falconer (1960). The number of sires and dams selected in each line differed among generations. The total number of chickens per generation varied from 116 animals in the 30th to 653 in the 18th generation in the low line and from 157 animals in the first to 633 in the 8th generation in the high line. If the number of animals from the first hatch was below 150 females, hatching was repeated up to 4 times. The number of the progeny could not have been predicted due to very erratic hatchability, especially in the line selected for high BW.

Random mating was used within each line. Chickens were reared collectively up to 8 wk of age in a windowless house with a deep litter system. After selection, parents were housed in separate pens within the same breeding house. Birds had free access to drinking water. During the growing period, chickens had ad libitum access to a diet containing 19% CP and 12 MJ of ME/kg. During the laying period, diet contained 17% CP and 11.7 MJ of ME/kg. The lighting regimen used was 23L:1D in wk 0 and then lighting was gradually decreased to 8L:16D in wk 2 to 18; then it was increased in weekly increments of 1.0 h until 15L:9D was reached.

Statistical Analyses

Changes in BW over generations were estimated by the mixed model:

$$y_{ijkl} = \mu + G_i + S_j + GS_{ij} + p_{ik} + e_{ijkl},$$

where y_{ijkl} represents the individual observation for BW, μ is the overall mean for BW, G_i is the fixed effect of the ith generation, S_j is the fixed effect of the jth sex, GS_{ij} is the effect of interaction between the ith generation and jth sex, p_{ik} is the random effect of the kth hatch group nested within the ith generation, and e_{ijkl} is the random error associated with the measurement of each individual. Least squares means (**lsmean**) were derived and were further analyzed using the NLIN procedure of SAS/STAT Statistical Software 9.2 (SAS Institute Inc., Cary, NC).

For fitting generation means against generation number or the cumulative selection differential, an exponential model (Herrendörfer and Bünger, 1988) was used:

$$y_i = a - (a - c)\exp^{\frac{-b*x_i}{(a-c)}} + e_i,$$

where y_i is the lsmean of BW, x_i is the generation, a is the theoretical selection limit, b is the maximal slope at $x = 0$, and c is the initial value. The same exponential model (Herrendörfer and Bünger, 1988) was also used to estimate genetic variance and realized heritability. When estimating realized heritability, y_i is the genetic response and x_i is the cumulative selection differential. Estimates of realized heritability for each generation were obtained from the first derivative of the exponential model. The half-life ($\mathbf{t_{1/2}}$) of the selection process was estimated as follows (Herrendörfer and Bünger, 1988):

$$t_{1/2} = \frac{(c - a)*\ln 0.5}{b}.$$

The estimates were expressed in units of generation or cumulative selection differential (in grams) as the time to reach half the limit. Intensity of selection was calculated as the ratio between selection differential and phenotypic SD.

Downloaded from http://ps.oxfordjournals.org/ at University of Bern on September 15, 2014

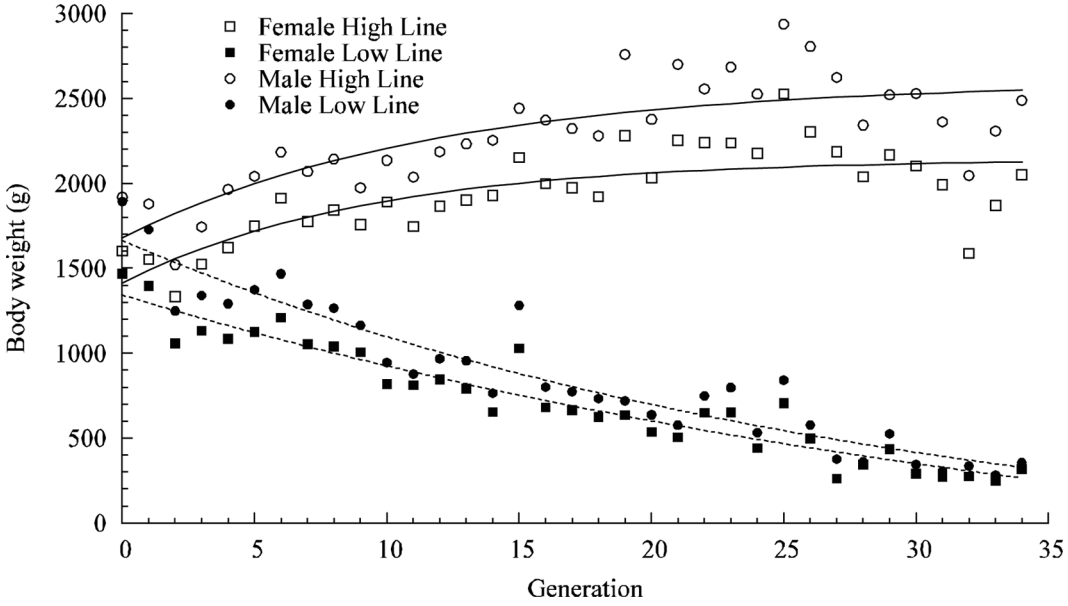


Figure 1. Least squares means of BW selected for 34 generations for high and low 8-wk BW.

RESULTS AND DISCUSSION

Response to Selection over Generations

Body weight changed considerably over the course of selection (Figure 1). Fluctuations in the response to selection were similar in both lines, except in the last 3 generations, where a decrease of response to selection was observed only in the high line. Initial responses were rapid and similar in both lines. In the first 10 generations, from 37.87% (females, low line) to 64.27% (females, high line) of the total response was achieved. The total response was higher in the low line. Comparison by sex revealed that selection was more effective for males, as expected according to the higher selection intensity (Figure 2). Meanwhile, selection after the 25th generation was still effective in the low line. The high line failed to respond and generation means decreased and became close to those in the 5th generation (Figure 1).

A reason for the episodic selection response could be the occurrence and fixation of a small number of mutations with large effects on BW. Dunnington and Siegel (1996) suggested 2 explanations for irregular response: a) sensitivity of genotypes to microenvironmental factors, or b) spontaneous mutations that may have occurred periodically. After 50 generations of divergent selection for 6-wk BW in a highly inbred mouse

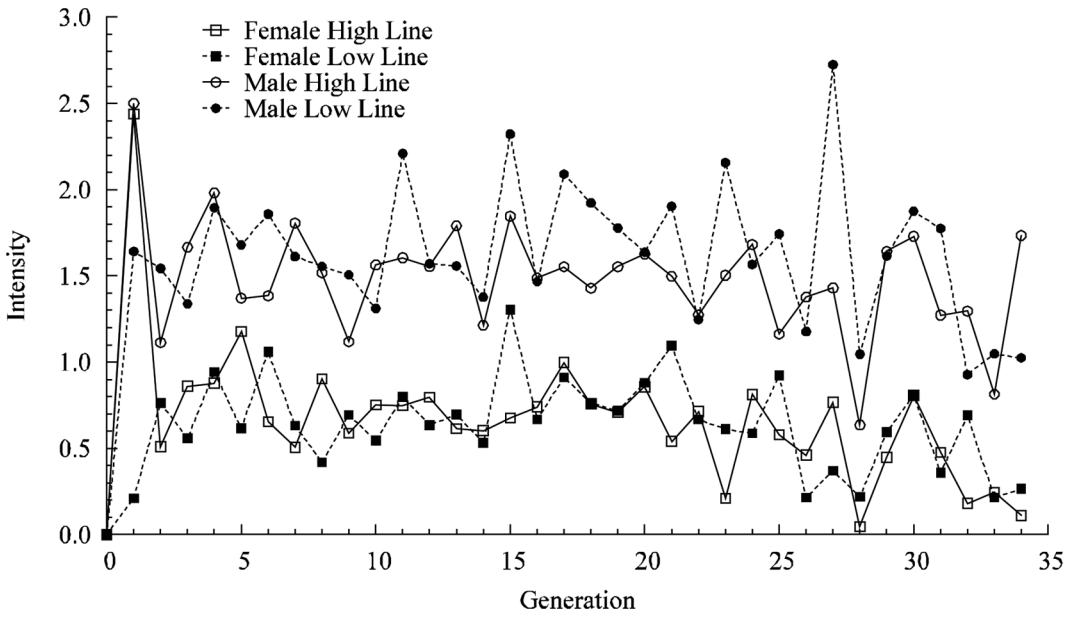


Figure 2. Intensity of selection over generations.

Downloaded from http://ps.oxfordjournals.org/ at University of Bern on September 15, 2014

Table 1. Parameter estimates for fitting the nonlinear model to generation means (g)

	Low line		High line	
Item	Female	Male	Female	Male
Parameter[1]				
a	−524.4 ± 620.0	−257.4 ± 455.5	2,144.1 ± 79.8	2,598.4 ± 126.3
b	−47.2 ± 9.3	−67.1 ± 14.2	79.1 ± 35.0	78.5 ± 32.3
c	1,342.2 ± 54.6	1,663.4 ± 77.5	1,412.5 ± 113.4	1,677.9 ± 121.3
Half-life[2]	27.4	19.8	6.4	8.1
RSD[3]	104.99	144.65	175.10	197.41
R^2	90.96	88.99	58.10	63.00

[1]a = theoretical selection limit; b = maximal slope at x = 0; c = initial value.

[2]Half-life = half-life of the selection process (generation).

[3]RSD = residual variance (g^2).

line, Keightley (1998) calculated an increase of 0.23 to 0.57% in BW heritability per generation from new mutations. The selection was more effective in the low line. According to Lyon and Searle (1989), greater response in the low line is expected because more mutations are known to reduce growth.

Fitting the generation means in the nonlinear model was better in the low line (Table 1), where R^2 was 90.96% for females and 88.99% for males. The model was less suitable in the high line (R^2 = 58.10% for females and 63.00% for males). Linear and quadratic regression explained a similar quantity of variability (results not shown), which suggests that all models would adequately describe the data.

Estimates derived from the nonlinear model suggested that selection limits had not been reached in the low line. The negative estimates with high SE indicated that a plateau had not been reached. The response could theoretically be continued, but selection in the low line was restricted due to negative consequences on reproduction and the biological limit of BW. Responses to selection for increased BW were better fitted by quadratic than linear regression for both sexes, which may be attributed to the decrease of BW in the last generations. Nevertheless, parameter estimates by the nonlinear model in the high line are more informative. The estimates for selection limit (parameter **a**) were 454.3 g greater for males than females (Table 1), with similar maximal slope in the first generation. The estimated maximal selection response at the beginning of the experiment (parameter **b**) was 78.5 g per generation in males and 79.1 g per generation in females in the high line. In the low line, the corresponding values were higher in males (−67.1 g per generation) than in females (−47.2 g per generation). As expected, the half-life was reached much earlier in the high line for both sexes because the selection limit had not yet been (mathematically) reached in the low line.

The initial BW (parameter **c**; Table 1) estimated in the high line was 1,677.9 g in males and 1,412.5 g in females. At the end of the selection experiment (generation 34), the BW increased by 920.5 g (54.8%) to 2,598.4 g in males and by 731.6 g (51.7%) to 2,144.1 g in females (Table 1). Females (1,342.2 g) as well as males (1,663.4 g) in the low lines were slightly lighter in the first generation. During the selection experiment, BW in the low line decreased to 355.7 ± 51.1 g in males, and to 318.0 ± 50.9 g in females in the 34th generation.

A similar long-term experiment was done in Virginia (Dunnington and Siegel, 1985; Liu et al., 1994; Dunnington and Siegel, 1996; Dunnington et al., 2013) where White Plymouth Rock chickens were selected for low and high BW. The initial population had lower BW than our foundation stock. The mean BW were 878 and 708 g for males and females, respectively (Dunnington et al., 2013). Response to selection showed a linear increase in the first 36 generations, with a change of about 22 g per generation for males and 17 g per generation for females in both lines (Liu et al., 1994). The response over 34 generations in our experiment was nonlinear. Dunnington et al. (2013) recently published results of the response over 54 generations in the Virginia experiment. They reported a significant quadratic response in the low line, but in the high BW line the increase remained linear. In the line selected for low BW, selection limits occurred at 173 g for males and 129 g for females. By a comparison of the selected lines with relaxed lines, they confirmed continued maintenance of genetic variance in the high BW line. The continued linear response was also attributed to occurrence of novel, beneficial mutations.

In a long-term selection experiment, exponential models for fitting generation means against generation were used by Bünger and Herrendörfer (1994), Bünger et al. (1998), and Bünger and Hill (1999). Bünger and Herrendörfer (1994) used the nonlinear model for analyzing data from a selection experiment where mice were selected over 84 generations for high BW at 60 d. The exponential model explained 92.3% of variability. Only one application of the exponential model could be found where the response decreased over generations, in which Bünger and Hill (1999) described response to selection for low and high BW over 53 generations in mice, but there is no information on how well the exponential model fitted the data.

In the present study, no control population was used. Therefore, selection response was measured by the dif-

Downloaded from http://ps.oxfordjournals.org/ at University of Bern on September 15, 2014

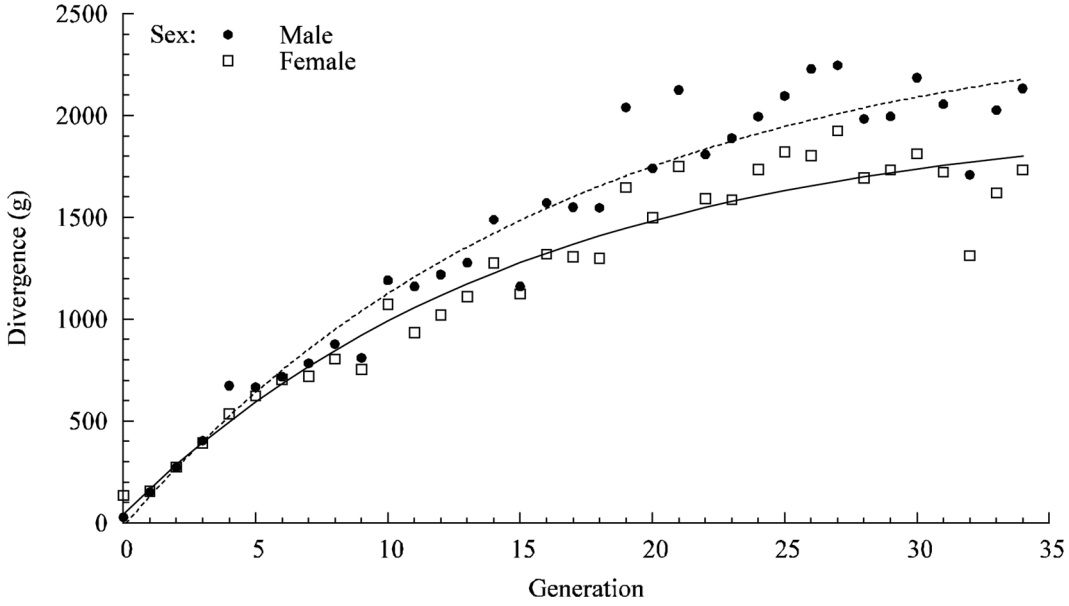


Figure 3. Difference of means 8-wk BW between the upward- and downward-selected lines.

ference between the high and the low lines (Figure 3). The difference between lines increased over generations in both sexes. The cumulative response over 34 generations, estimated by the nonlinear model, amounted to 2,178.1 g for males and 1,799.9 g for females. After 34 generations the lines differed by around 2,220 g for males and 1,860 g for females. The divergence increased up to the 27th generation. In later generations, there was a fluctuation around the plateau, or even a slight reduction between generation means in the low and high lines.

Intensity of Selection

Intensity of selection, expressed as the ratio between realized selection differential and phenotypic SD, was comparable between lines but differed significantly between sexes (Figure 2), as expected. The proportion of males selected to be sires of the next generation was much smaller than the proportion of females selected. The proportions between lines were kept as close as possible. The reduction over generations was significant in females but not in males. Thus, selection intensity in females declined more or less constantly at the rate of -0.015 ± 0.004 per generation in the high line and -0.010 ± 0.004 in the low line. It was relatively constant over generations in males; linear regression coefficients were not significantly different from zero. An evident decrease of selection intensity was obtained after the 20th generation due to reproductive problems and the consequent deficiency of birds available. Liu et al. (1994) reported a decreasing selection intensity over generations in the line selected for increased BW and an increasing one in the line selected for low BW in chickens. In the high line, linear regression coefficients were -0.015 for males and -0.014 for females, whereas in the low line increases of 0.015 for males and 0.014 for females were obtained.

Response Versus Cumulative Selection Differential

Selection response depends on selection differential (Figure 4). Plotting response against cumulative selection differential eliminates the differences of selection differential over generations (Falconer, 1955). The larger selection differential was possible in males due to the smaller number of animals needed. The high line had a larger cumulative selection differential than the low line. Regressions of diminishing responses were well fitted in the nonlinear model (Table 2). Parameter b represents an estimate of the initial realized heritability. Initial heritability was lower in males.

Variability

Phenotypic selection is expected to reduce genetic variance. However, the genetic SD shows a linear decrease over generations (Figure 5). The reduction of genetic variation is expected and in agreement with basic theory in quantitative genetics (Robertson, 1960). The reduction was -1.42 g per generation for females and -2.22 g for males. Results were consistent with earlier reports of declining genetic variation over generations (Marks, 1978; Bünger and Herrendörfer, 1994).

Selection for high BW resulted in negligible positive trends of increased phenotypic SD (Figure 6). The change per generation in the high line was 1.0 g for males ($P = 0.0695$) and 1.8 g for females ($P = 0.0032$). In the low line, phenotypic SD was constant from the initial to the 22nd generation. Later on, the regression coefficient was -4.3 ± 0.9 g per generation for females

Downloaded from http://ps.oxfordjournals.org/ at University of Bern on September 15, 2014

Table 2. Parameter estimates for fitting nonlinear model to selection response (g) against cumulative selection differential (g)

	Low line		High line	
Item	Female	Male	Female	Male
Parameter[1]				
a	−3,056.2 ± 2,829.0	−2,773.5 ± 1,283.8	582.2 ± 114.6	698.4 ± 144.9
b	−0.47 ± 0.12	−0.23 ± 0.06	0.55 ± 0.26	0.24 ± 0.10
c	−162.5 ± 58.4	−255.8 ± 80.1	−217.5 ± 129.2	−240.6 ± 122.0
RSD[2]	116.28	150.76	176.86	196.91
R^2	88.91	88.04	57.25	63.19

[1]a = theoretical selection limit; b = maximal slope at x = 0, estimate of initial realized heritability; c = initial value.
[2]RSD = residual variance (g^2).

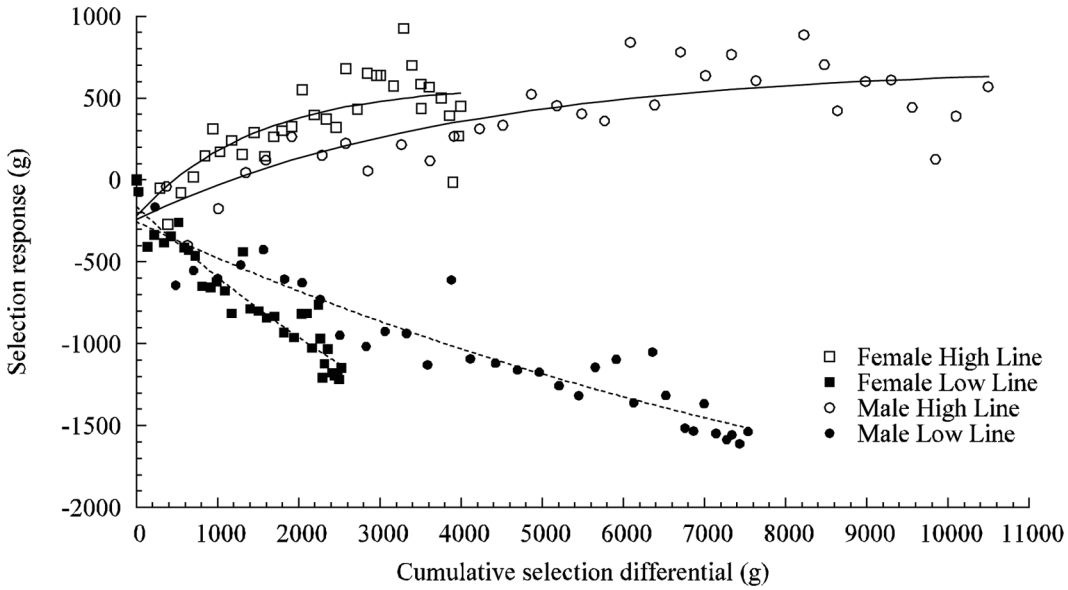


Figure 4. Cumulative selection response versus cumulative selection differential.

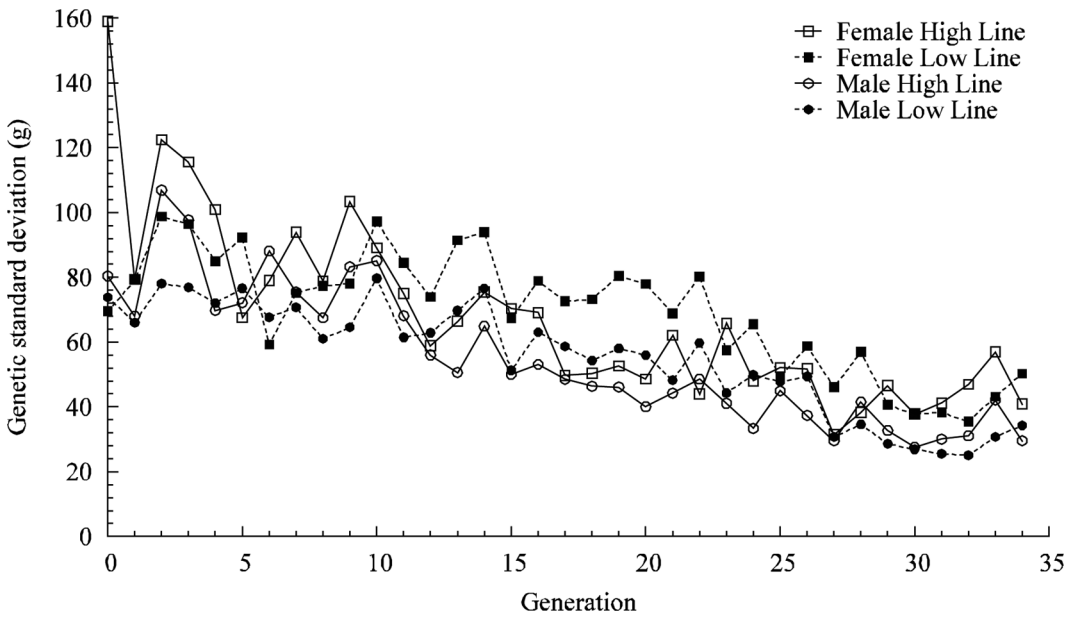


Figure 5. Genetic SD over generations.

Downloaded from http://ps.oxfordjournals.org/ at University of Bern on September 15, 2014

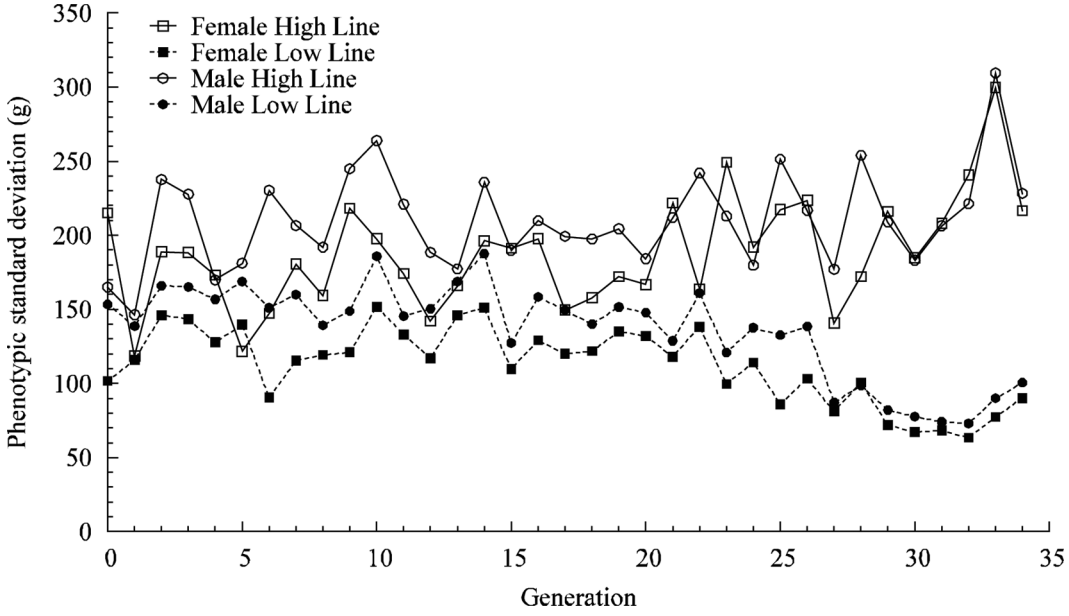


Figure 6. Phenotypic SD over generations.

and -5.4 ± 1.1 g per generation for males in the low line. Falconer and Mackay (1996) named such changes the scale effect, because phenotypic variance changes with population mean, and suggested comparing variability on a logarithmic scale or by the CV. The CV increased in the low line with a change of 0.485% per generation for males and 0.530% for males. In the high line, CV were constant for both sexes. Considering that phenotypic variance is a sum of genetic and environmental components, the proportion of environmental variance (e^2) can be expressed as $1 - h^2$. Environmental variance increased over the selection experiment in both lines, even more in the high line, probably due to increased environmental sensitivity.

Realized Heritability

Analyses of each line separately revealed differences in heritabilities between the lines (Figure 7). The estimated realized heritability decreased over generations. During the selection experiment, realized heritability declined from 0.24 to 0.02 in males and from 0.55 to 0.04 in females of the high line. The corresponding decreases in the low line were from 0.23 to 0.12 in males and from 0.47 to 0.31 in females (Figure 7). After 84 generations of selection for BW in mice, Bünger and Herrendörfer (1994) found that realized heritability declined from a high (0.361) to a low (0.0004) value. Heritability declined more rapidly in the high line than in the low line. The genetic SD decreased similarly in both lines (Figure 5), whereas the phenotypic SD increased in the high line and was constant in the low line during the first 22 generations. After the 22nd generation, however, an increase was obtained in the high line and a decrease in the low line (Figure 6). According to the higher estimates of realized heritability in the 34th generation, it seems that the low line did not reach a plateau.

Comparison of realized heritabilities between studies can be complicated by the number of generations under selection. However, it is possible to summarize selection studies in which selection has been practiced for an approximately equal number of generations. Heritabilities were consistently higher in females than in males, which is in contrast with the results of Liu et al. (1994). Realized heritabilities in their study were calculated as regression of cumulative responses on cumulative selection differentials and the ratio between total response in the last generation and total selection differential in the same generation. Realized heritabilities for males and females adjusted for the number of progeny ranged from 0.22 to 0.28 for the high line and from 0.23 to 0.28 for the low line. Although in the high line heritabilities were greater in the interval from the first to the 18th generations, in the low line heritabilities were greater from the 18th to the 36th generations, which is in con-

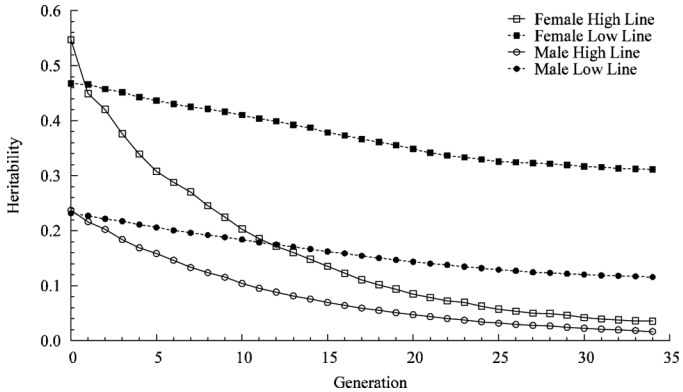


Figure 7. Realized heritabilities over generations.

Downloaded from http://ps.oxfordjournals.org/ at University of Bern on September 15, 2014

flict with the prediction of declining heritability over generations. In theory, selection for traits with moderate to high heritabilities would narrow genetic variation and therefore would lead to a reduction in heritability (Falconer and Mackay, 1996). Siegel (1962) summarized 176 published heritability estimates for BW of chickens from 6 to 12 wk of age. The average heritability was 0.41, ranging from 0.29 to 0.54. However, BW is a trait with moderate heritability, which depends on population characteristics and on selection to change the population. Selection in 2 directions yielded very different realized heritabilities. According to Falconer and Mackay (1996), each is a valid description of the response but for several reasons (e.g., systematic changes due to environmental trends or inbreeding depression, random drift, and so on) they cannot both be valid estimates of the heritability in the base population.

ACKNOWLEDGMENTS

The authors gratefully acknowledge the contributions of colleagues to the careful breeding of the experimental lines of chickens. We are grateful to A. R. Byrne (retired, Jožef Stefan Institute, Ljubljana, Slovenia) for help with editing. The constructive comments of the reviewers are also acknowledged.

REFERENCES

Barbato, G. F. 1992. Divergent selection for exponential growth rate at fourteen and forty-two days of age. 1. Early responses. Poult. Sci. 71:1985–1993.

Bünger, L., and G. Herrendörfer. 1994. Analysis of a long-term selection experiment with an exponential model. J. Anim. Breed. Genet. 111:1–13.

Bünger, L., and W. G. Hill. 1999. Inbred lines of mice derived from long-term selection on fat content and body weight. Mamm. Genome 10:645–648.

Bünger, L., U. Renne, G. Dietl, and S. Kuhla. 1998. Long-term selection for protein amount over 70 generations in mice. Genet. Res. 72:93–109.

Dahmane Gošnak, R., I. Eržen, A. Holcman, and D. Škorjanc. 2010. Effects of divergent selection for 8-week body weight on postnatal enzyme activity pattern of 3 fiber types in fast muscles of male broilers (*Gallus gallus domesticus*). Poult. Sci. 89:2651–2659.

Dempster, E. R., I. M. Lerner, and D. C. Lowry. 1952. Continuous selection for egg production in poultry. Genetics 37:693–708.

Dunnington, E. A., C. F. Honaker, M. L. McGilliard, and P. B. Siegel. 2013. Phenotypic responses of chickens to long-term, bidirectional selection for juvenile body weight—Historical perspective. Poult. Sci. 92:1724–1734.

Dunnington, E. A., and P. B. Siegel. 1985. Long-term selection for 8-week body weight in chickens—Direct and correlated responses. Theor. Appl. Genet. 71:305–313.

Dunnington, E. A., and P. B. Siegel. 1996. Long-term divergent selection for eight-week body weight in White Plymouth Rock chickens. Poult. Sci. 75:1168–1179.

Eisen, E. J. 1980. Conclusions from the long-term selection experiment with mice. Z. Tierzuchtg. Zuchtbiol. 97:305–319.

Falconer, D. S. 1955. Patterns of response in selection experiments with mice. Cold Spring Harb. Symp. Quant. Biol. 20:178–196.

Falconer, D. S. 1960. Introduction of Quantitative Genetics. 1st ed. Oliver and Boyd Ltd., Edinburgh and London, UK.

Falconer, D. S., and T. F. C. Mackay. 1996. Introduction of Quantitative Genetics. 4th ed. Longman Group, Harlow, Essex, UK.

Frahm, R. R., and K. I. Kojima. 1966. Comparison of selection responses on body weight under divergent larval density condition in *Drosophila pseudoobscura*. Genetics 54:625–637.

Harris, D. L. 1982. Relation to breeding populations size intensity and accuracy with additive gene action. Genetics 100:511–532.

Herrendörfer, G., and L. Bünger. 1988. Estimation of the h2-function in long term selection experiments. Probleme der angewandten Statistik, Heft, 26, 45–49. Proc. Intern. Conf. Population Mathematics: Proc. Intern. Conf. Population Mathematics: Oct. 25–31., 1987, Schwerin, Germany.

James, J. W. 1965. Response curves in selection experiments. Heredity 20:57–63.

Keightley, P. D. 1998. Genetic basis of response to 50 generations of selection on body weight in inbred mice. Genetics 148:1931–1939.

Liu, G., E. A. Dunnington, and P. B. Siegel. 1994. Responses to long-term divergent selection for eight-week body weight in chickens. Poult. Sci. 73:1642–1650.

Lyon, M., and A. Searle. 1989. Genetic Variants and Strains of the Laboratory Mice. 2nd ed. Oxford University Press, Oxford, UK.

Marks, H. L. 1978. Long-term selection for four-week body weight in Japanese quail under different nutritional environments. Theor. Appl. Genet. 52:105–111.

Márquez, G. C., P. B. Siegel, and R. M. Lewis. 2010. Genetic diversity and population structure in lines of chickens divergently selected for high and low 8-week body weight. Poult. Sci. 89:2580–2588.

Richardson, R. H., K. Kojima, and H. L. Lucas. 1968. An analysis of short term selection experiments. Heredity 23:493–506.

Robertson, A. 1960. A theory of limits in artificial selection. Proc. R. Soc. Lond. B Biol. Sci. 153:234–249.

Siegel, P. B. 1962. Selection for body weight at 8 weeks of age. 1. Short time response and heritabilities. Poult. Sci. 41:141–145.

Terčič, D., and A. Holcman. 2008. Long–term divergent selection for 8-week body weight in chickens—A review of experiments. Acta Agric. Slov. 92:131–138.

Terčič, D., A. Holcman, P. Dovč, D. R. Morrice, D. Burt, P. M. Hocking, and S. Horvat. 2009. Identification of chromosomal regions associated with growth and carcass traits in an F_3 full sib intercross line originating from a cross of chicken lines divergently selected on body weight. Anim. Genet. 40:743–748.

Downloaded from http://ps.oxfordjournals.org/ at University of Bern on September 15, 2014